Communication Lower Bounds for Multi-Party Graph Traversal

Quarter 2 Project Report

Ryan Batubara `rbatubara@ucsd.edu`

Ivy Hawks `mahawks@ucsd.edu`

Darren Ho `dah103@ucsd.edu`

Ciro Zhang `ciz001@ucsd.edu`

Mentor: Dr. Shachar Lovett `slovett@ucsd.edu`

Abstract

TBD

Repository: `https://github.com/rybplayer/DSCCapstone`

1 Introduction

1.1 Lower bounds and Communication Complexity

As we gain and store every more information, we constantly need more effective ways to process and store that information. To do this more effectively, we can either create more powerful hardware, or create ever more efficient algorithms. We will consider the latter course. Many algorithm designers have created enormously efficient algorithms for solving many core problems in querying and optimization, but improvement does beg the question–how fast can we make these algorithms? Certainly, asked to check if two numbers are the same, our natural inclination might be to simply compare all of the digits, resulting in a protocol running in $\mathcal{O}(n)$ time. However, is it possible to do better than this? And can we prove it? While classical computing might provide tools for the case of equality—No, we cannot do better—many more complex questions such as optimization using linear programs cannot be proven using these classical tools. Communication Complexity provides a model under which we can transform these logical processes, for which we lack adequate mathematical tools to prove bounds on, into combinatorial ones for which we can find solutions much more easily.

1.2 Motivation from multi-party traversal case studies

Large-scale transportation and infrastructure disruptions create a natural model for multi-party graph traversal. Consider the 2010 European volcanic ash shutdown: air-navigation service providers, airports, and airlines each control disjoint portions of airspace and schedules. Rerouting passengers is a shortest-path problem in a layered graph whose feasible vertices and edges change over time. No party can broadcast its full real-time state, both for operational and privacy reasons, so route feasibility must be computed with limited information exchange.

Similar graph-traversal constraints appear in urban transit shutdowns and phased restorations, where distinct agencies own stations and tunnels. After Hurricane Sandy, service availability evolved over weeks, and agencies coordinated alternate routes under partial knowledge of what infrastructure was open. Highway detours after the I-95 bridge collapse in Philadelphia illustrate the same phenomenon in a road network: different operators own highway segments and toll roads, and a single failed cut edge forces traffic onto alternative paths with asymmetric knowledge of capacities.

The maritime and logistics examples are equally instructive. The 2021 Suez Canal blockage forced carriers to reroute around the Cape of Good Hope or wait, a decision that depends on schedule and capacity information held privately by shipping lines and port operators. When the Baltimore Key Bridge collapsed, access to the port was blocked and cargo was diverted to alternate ports, again requiring coordination across a sea–port–land graph with private operational constraints.

Communication bottlenecks become even sharper in critical infrastructure networks. During the 2021 Texas ERCOT blackout, grid operators and utilities effectively solve a constrained flow and connectivity problem as generation and transmission nodes go offline. The Colonial Pipeline cyber disruption similarly forced fuel logistics to reroute through alternate depots and terminals with incomplete visibility. Finally, the 2021 Meta backbone withdrawal and the 2016 Dyn DNS attack show that digital services rely on reachability in routing and dependency graphs, where policy and security constraints limit information sharing across administrative domains.

These cases motivate a theoretical study of multi-party graph traversal: agents must collaboratively decide reachability and shortest paths while their private subgraphs, capacities, or weights are not fully shareable. The fundamental question is how much communication is necessary, and how randomized protocols, sketches, or partial disclosures can reduce the burden while preserving correctness.

1.3 Why communication is the bottleneck

In the model we consider, each party has unlimited local computation but limited ability to share data. This is realistic: air traffic control centers, network operators, or utilities can compute detailed local routes, but cannot afford to broadcast all local states and constraints due to bandwidth, privacy, regulatory, or security limitations. The communication cost, not the local computation, is what constrains real-time coordination.

Communication complexity makes this separation explicit. In particular, it shows that even when each party can compute arbitrarily powerful local summaries, some global questions still require a large number of bits to be exchanged. For example, deciding whether two privately held sets intersect already requires linear communication in the worst case. This lower bound appears again in many traversal problems via reductions that embed set disjointness or equality into graph reachability. Thus, communication is not simply a practical nuisance: it is an inherent barrier that must be designed around.

1.4 Why bounds matter for hardware, networks, and algorithms

Lower and upper bounds provide practical guidance in three ways. First, they inform hardware design. If a reachability or shortest-path query provably requires $\Omega(n)$ bits of communication, then any system intended to answer such queries at scale must provision sufficient bandwidth or memory to move that information. Second, bounds guide network architecture: if communication is unavoidable, the topology of communication links, the placement of caches, and the choice of aggregation points become part of the algorithmic design. Third, bounds shape algorithm development: a tight lower bound rules out overly ambitious protocols and motivates either approximate solutions, randomized protocols with small error, or problem relaxations that admit sublinear communication. Conversely, upper bounds show which tasks are feasible under realistic constraints, making them valuable to practitioners who must trade correctness, latency, and privacy.

1.5 Communication complexity basics

We briefly formalize the model. Let $f : \mathscr{X} \times \mathscr{Y} \to \{0,1\}$ be a function. In the two-party model, Alice receives $x \in \mathscr{X}$ and Bob receives $y \in \mathscr{Y}$. A deterministic protocol is a binary decision tree whose nodes are labeled by which party speaks and whose edges are labeled by bits; the transcript is the sequence of bits exchanged. The protocol must output $f(x, y)$ at a leaf. The deterministic communication complexity $D(f)$ is the minimum, over all correct protocols, of the maximum transcript length over all inputs.

Randomized protocols allow parties to use random coins to reduce communication. A public-coin protocol assumes a shared random string, while a private-coin protocol lets each party sample independently. Fixing the randomness yields a deterministic protocol, so a randomized protocol is a distribution over deterministic protocols. The randomized communication complexity $R_\varepsilon(f)$ is the minimum number of communicated bits needed to compute f with error at most ε on every input. Public-coin and private-coin models are equivalent up to small overhead, and randomness often trades a small probability of error for large savings in communication.

Two classical benchmark problems are equality and disjointness. Equality asks whether $x = y$ for $x, y \in \{0,1\}^n$, and disjointness asks whether $X \cap Y = \emptyset$ for subsets of $[n]$. Both are complete for a wide range of lower-bound techniques. Deterministically, equality requires n bits in the worst case, and disjointness requires $\Omega(n)$ communication. Randomization yields large gains for equality via hashing, but not for disjointness, which remains hard.

1.5.1 Proof Example: Equality

Here we will detail a small example proof of a well known problem in the field: That of equality. While the proof itself is simple, the crucial point of it is that it is valid independent of any specific protocol. Thus, rather than showing that any specific protocol will not perform better than this, it shows that no protocol can exist for this problem which will do any better.

This proof utilizes the concept of a matrix for a function, and a monochromatic rectangle. A functions communication matrix can be thought of as a table in which the columns correspond to one parties input, and the rows correspond to the other parties input. Each entry, M_x, y represents the output of the function $f(x, y)$. A monochromatic rectangle in this matrix is a rectangle where all input pairs produce the same output, allowing for protocols which can halt before transmitting their entire input if they can identify that their input falls in a large one. Conversely, in order to halt every time, the protocol must in the worst case transmit enough bits to find the smallest rectangle, and therefore finding a bound on the number of monochromatic rectangles for a given output can induce a bound for any protocol which computes this function.

The Deterministic communication complexity for *Equality* is $\Omega(n)$

Proof. We observe that the matrix for E.Q. does not contain any large 1-monochromatic rectangles. Given two points, (x, x') where $x \neq x'$, (x, x) and (x', x') cannot be in the same monochromatic rectangles, as otherwise (x, x') and (x', x) would also occupy this same rectangle. Thus, we have that there must be at least n monochromatic rectangles.

In this report, we focus on studying the specific case of graph problems, in which two or more parities are given a subset of the graph and asked to compute its properties. Primarily we research vertex reachability, that is, given two vertices (u, v), identify it there is a path between them which uses only the subsets of the graph owned by the communicating parties.

2 The Traversal Problem

In this section we will explore our findings from our research.

2.1 Problem

Consider a graph G with edges E and vertices V, and subgraphs $G_1, \dots, G_k \subset G$. Let $G_i = (E_i, V_i)$ for these subgraphs. For now let us suppose the graph is unweighted and undirected. The subgraphs need not cover G and can overlap. Each player i knows G and G_i, but not the other player's graphs. (This is akin to a map of a city, where each player is a different regional taxi company.) The goal is, given some starting vertex $v \in V$, and some end vertex, $w \in V$ to:

1. Identify if it is possible to construct a path $v, v_1, \dots, v_l, w$ such that every edge on the path is in $\bigcup_{i=0}^{k} E_i$.
2. If such a path exists, find the minimum number of edges needed to get from v to w, while using only edges belonging to $\bigcup_{i=0}^{k} E_i$.

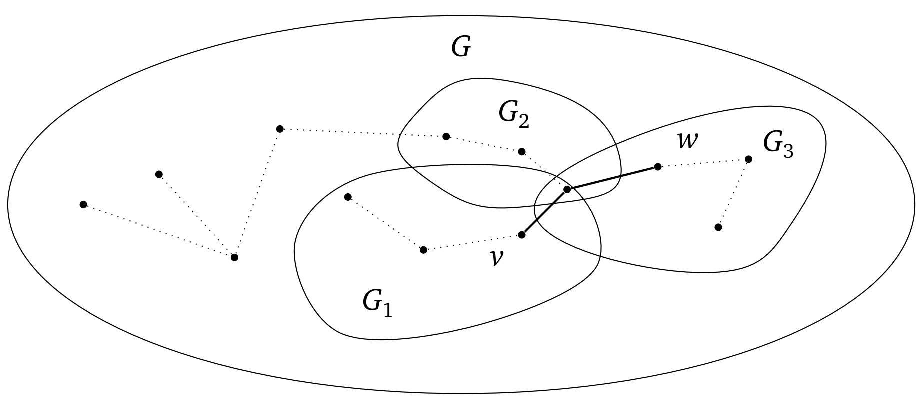

Figure 1: A valid path from v to w in bold. Note that G may contain nodes not belonging to any player.

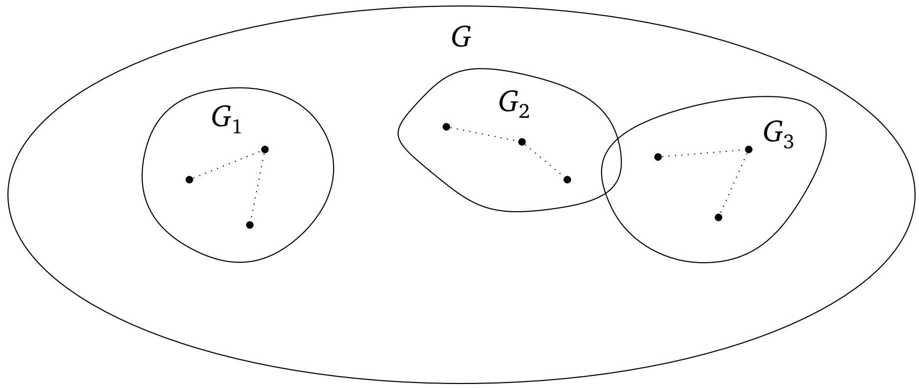

Figure 2: Worst case for item 1: Disjoint Subsets

2.2 Lower bound using Equality

Equality problem. In the two-party Equality problem, Alice is given a string $x \in \{0,1\}^n$ and Bob is given a string $y \in \{0,1\}^n$. The goal is to determine whether $x = y$. It is known that Equality has deterministic communication complexity $\Omega(n)$.

Equality gadget. We construct a gadget that enforces equality at a single bit position. The gadget is shown in Figure 3. The gadget consists of two parallel branches.

1. The top branch corresponds to the assignment $x_i = y_i = 1$,
2. The bottom branch corresponds to the assignment $x_i = y_i = 0$.

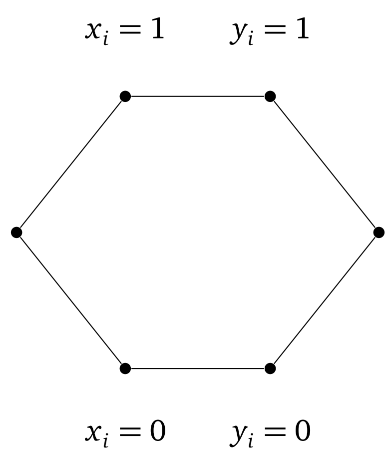

Figure 3: Gadget for proving equality. A path only exists if $x_i = y_i$ for either branch.

Alice enables exactly one entry node of the gadget depending on x_i, and Bob enables exactly one exit node depending on y_i. A path exists through the gadget if and only if Alice and Bob select nodes on the same branch, which holds exactly when $x_i = y_i$.

Reduction to our graph problem. We construct a graph by chaining one copy of this gadget for each index $i \in [n]$ between a global start vertex v and end vertex w. In the resulting graph, there exists a path from v to w if and only if all gadgets are traversable, which occurs if and only if $x_i = y_i$ for all i, i.e., if and only if $x = y$.

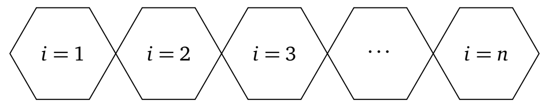

Figure 4: Equality gadgets chained as adjacent hexagonal modules.

Therefore, any protocol that solves the reachability problem on this graph can be used to solve the Equality problem. Since Equality requires $\Omega(n)$ deterministic communication, it follows that our graph problem also requires $\Omega(n)$ deterministic communication.

2.3 Lower bound using Disjointness

Set Disjointness problem. Alice is given a vector $x \in \{0,1\}^n$ and Bob is given a vector $y \in \{0,1\}^n$. The goal is to determine whether the sets represented by x and y are disjoint. It is known that Disjointness has deterministic communication complexity $\Omega(n)$.

Disjointness gadget. We construct a gadget that enforces disjointness at a single bit position. The gadget is shown in Figure 5. The gadget is constructed so that traversal is possible for all assignments except when both parties contain the element.

1. The top branch corresponds to when x_i has the element but y_i does not
2. The middle branch corresponds to when x_i and y_i do not have the element
3. The top branch corresponds to when y_i has the element but x_i does not

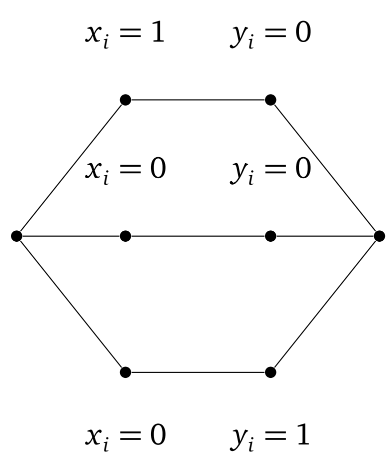

Figure 5: Gadget for proving equality, a path only exists if x_i and y_i do not have the same element

Reduction to our graph problem. We construct a graph by chaining one copy of this gadget for each index $i \in [n]$ between a global start vertex v and end vertex w. In the resulting graph, there exists a path from v to w if and only if all gadgets are traversable, which occurs if and only if x and y does not contain the same element

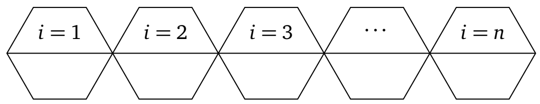

Figure 6: Equality gadgets chained as adjacent hexagonal modules.

Therefore, any protocol that solves the reachability problem on this graph can be used to solve the Disjoint problem. Since Disjoint requires $\Omega(n)$ deterministic communication, it follows that our graph problem also requires $\Omega(n)$ deterministic communication.

3 Other Problems

1. Unbounded Traversal with Common Input ("0-communication" triviality). Model. Fix any class of finite graphs $\mathcal{G}$ and any set of queries $\mathcal{Q}$. An instance is a pair (G,q) with $G \in \mathcal{G}$ and $q \in \mathcal{Q}$. Let $\mathsf{Ans} : \mathcal{G} \times \mathcal{Q} \to \{0,1\}$ be any decision predicate.

Consider a two-party communication problem in which Alice's input is $x = (G,q)$ and Bob's input is $y = (G,q)$ *identically the same string*. (Equivalently, $y = x$ is promised.)

Define the Boolean function

$$f_{\text{common}}(x,y) \;=\; \mathsf{Ans}(G,q) \quad \text{under the promise } x = y = (G,q).$$

Claim (Triviality). $D(f_{\text{common}}) = 0$ and $N^1(f_{\text{common}}) = N^0(f_{\text{common}}) = 0$.

Proof. Since $x = y$, both parties can compute $\mathsf{Ans}(G,q)$ locally with unbounded computation. Therefore a deterministic protocol that sends no bits exists: each outputs $\mathsf{Ans}(G,q)$. Hence $D(f_{\text{common}}) \le 0$, so $D(f_{\text{common}}) = 0$.

For nondeterminism, a nondeterministic protocol may also send no bits: each party ignores the certificate and outputs $\mathsf{Ans}(G,q)$. Thus $N^1(f_{\text{common}}) = N^0(f_{\text{common}}) = 0$.

Lower bound. Communication complexity is always ≥ 0 by definition, so the upper bound is tight.

Embedding note. Any nontrivial lower bound requires *private information* (e.g., Alice and Bob see different parts of G, or have different start locations/vision transcripts). Without private inputs, f collapses to a one-input problem, and communication is identically 0.

2. Black Hole Search on a Path (communication $\Theta(\log n)$). Goal. Formalize a decision/identification task where the *only* uncertainty is the black-hole location, and determine the deterministic and nondeterministic communication needed.

Instance. Let P_n be the path graph on vertices $\{1, 2, \ldots, n\}$ with edges $(i, i+1)$. There is an unknown black-hole vertex $b \in [n]$. A probe (agent) that *visits* b is destroyed and sends no further messages. If it never visits b, it survives and may report back.

Communication model. Alice and Bob cooperate and can launch probes from vertex 1. A probe may walk deterministically along the path; upon returning to 1 alive, it can send a *single bit* "alive" to both parties. (If destroyed, no bit is sent.) The *communication cost* is the total number of bits ever successfully sent by surviving probes.

Problem (BH-LOC). The output is the *exact* index $b \in [n]$. Equivalently, define f_b as the function mapping the unknown b to its binary encoding, and ask the parties to output b.

Claim. The deterministic communication complexity to identify b satisfies

$$D(\text{BH-LOC}_n) = \Theta(\log n).$$

Upper bound (protocol achieving $O(\log n)$ bits). Maintain an interval $I = [L, R]$ known to contain b (initially $[1, n]$). Repeat while $L < R$: let $m = \lfloor \frac{L+R}{2} \rfloor$. Launch a probe that walks from 1 to m and then returns to 1 along the same path.

- If the probe returns alive, then it never visited b, so $b \notin [1, m]$ and hence $b \in [m+1, R]$. Update $L \leftarrow m+1$.
- If the probe is destroyed, it must have visited b on the way to m, so $b \in [L, m]$. Update $R \leftarrow m$.

Each iteration produces at most one bit of communication (alive/no-alive, where "no-alive" is inferred from silence). The interval size halves each time, so there are at most $\lceil \log_2 n \rceil$ iterations and thus $O(\log n)$ communicated bits. When $L = R$, output $b = L$.

Lower bound (why $\Omega(\log n)$ bits are necessary). Any correct deterministic protocol must distinguish among n possible black-hole locations. Each bit of communication can create at most a factor-2 branching in the transcript. If at most c bits are communicated, there are at most 2^c possible transcripts, hence at most 2^c different outputs can be forced. Correctness for all $b \in [n]$ requires $2^c \geq n$, so $c \geq \lceil \log_2 n \rceil$. Thus $D(\text{BH-LOC}_n) \geq \log_2 n$, matching the protocol up to constants.

Nondeterministic complexity. To certify that the black hole is at location b, a prover can send the index b using $\lceil \log_2 n \rceil$ bits, after which the parties verify by running the above test specialized to b (e.g., test $m = b$ and $m = b-1$). Hence $N^1(\text{BH-LOC}_n) = \Theta(\log n)$. Similarly N^0 is also $\Theta(\log n)$ since the output is an index among n possibilities.

Embedding used for the lower bound. This lower bound is the standard *information* bound: distinguishing n cases requires $\Omega(\log n)$ bits. Equivalently, one can embed the INDEX function by letting b encode an $\log n$-bit string.

3. Graph Isomorphism as Equality under a Rigidity Promise (communication $\Theta(n^2)$).
Inputs. Fix n. Alice receives an n-vertex labeled simple graph G via its adjacency matrix $A \in \{0,1\}^{n \times n}$ (with zeros on the diagonal and symmetry). Bob receives an n-vertex labeled simple graph H via adjacency matrix $B \in \{0,1\}^{n \times n}$. Let $\text{GI}_n(G, H) = 1$ iff G and H are isomorphic (i.e., $A = PBP^\top$ for some permutation matrix P).

Why unrestricted GI is not "just equality". Without restrictions, $\mathrm{GI}_n(G,H)=1$ may hold even when $A \neq B$ as strings, so it is *not* literally the Equality function on the adjacency matrices.

Promise family (Rigid graphs). Let $\mathscr{R}_n$ be any family of labeled graphs on $[n]$ with *trivial automorphism group*: for every $G \in \mathscr{R}_n$, the only permutation π with $\pi(G) = G$ is the identity. (Equivalently: G has no nontrivial relabeling that preserves adjacency.)

Define the *promised* problem:

$$\mathrm{GI}_n^{\mathscr{R}}(G,H) = \mathrm{GI}_n(G,H) \quad \text{with the promise } G,H \in \mathscr{R}_n.$$

Claim (Triviality-to-equality under the promise). On $\mathscr{R}_n \times \mathscr{R}_n$, isomorphism equals literal equality:

$$\forall G,H \in \mathscr{R}_n, \qquad \mathrm{GI}_n(G,H) = 1 \iff G = H \text{ (as labeled graphs)}.$$

Proof. If $G = H$ as labeled graphs, then clearly they are isomorphic (identity permutation), so $\Rightarrow$ holds. Conversely, suppose $\mathrm{GI}_n(G,H) = 1$. Then there exists a permutation π with $\pi(H) = G$. Apply π^{-1} to both sides: $H = \pi^{-1}(G)$, so G is isomorphic to itself via $\pi \circ \sigma$ for some σ mapping G to H; more simply, since G and H are isomorphic, there exists a bijection φ from $V(G)$ to itself that preserves adjacency between G and G when composing the two isomorphisms $G \to H$ and $H \to G$. This yields a graph automorphism of G. Rigidity of G forces this automorphism to be identity, hence the original isomorphism must be identity on labels, so $G = H$ as labeled graphs.

Deterministic communication complexity. Let $m = \binom{n}{2}$ be the number of independent adjacency bits. Then, under the promise, $\mathrm{GI}_n^{\mathscr{R}}$ is exactly EQ_m (equality on m bits). Therefore

$$D(\mathrm{GI}_n^{\mathscr{R}}) = \Theta(m) = \Theta(n^2).$$

Upper bound protocol (achieves $O(n^2)$). Alice sends her adjacency matrix (or just the upper triangle) to Bob using m bits. Bob checks if his matrix equals Alice's; if yes output 1, else 0. Cost $m = \Theta(n^2)$ bits.

Lower bound (embedding of EQ_m). Equality on m bits has deterministic communication complexity $D(\mathrm{EQ}_m) = m$: Alice must reveal enough to distinguish her m-bit string from all others. Formally, if fewer than m bits are sent, by pigeonhole there exist distinct $x \neq x'$ producing the same transcript, and Bob cannot distinguish (x,x) from (x',x), causing an error on equality. Since $\mathrm{GI}_n^{\mathscr{R}}$ equals EQ_m under the promise, the same lower bound applies.

Nondeterministic bounds. For equality:

$$\begin{aligned} N^1(\mathrm{EQ}_m) &= \Theta(m) && \text{(certificate is the full string)}, \\ N^0(\mathrm{EQ}_m) &= \Theta(\log m) && \text{(certificate is an index of a differing bit)}. \end{aligned}$$

Thus

$$N^1(\mathrm{GI}_n^{\mathscr{R}}) = \Theta(n^2), \qquad N^0(\mathrm{GI}_n^{\mathscr{R}}) = \Theta(\log n).$$

Problems embedded to prove bounds. The tight deterministic bound uses an embedding of EQ_m. The nondeterministic 0-certificate bound uses embedding of NEQ_m via an index witness.

4. Multi-agent Sokoban with Shared Instance (communication 0). Sokoban instance. Fix a grid $[W]\times[H]$ with walls $S \subseteq [W]\times[H]$. There are k labeled movable blocks with initial positions $B_1,\dots,B_k \in ([W]\times[H])\setminus S$ and k labeled goal cells $G_1,\dots,G_k \in ([W]\times[H])\setminus S$. There are k agents with initial positions $A_1,\dots,A_k$. A move allows an agent to step to a neighboring non-wall cell; if it steps into a block cell, it may push the block one step forward provided the destination cell is empty and non-wall. (Any agent may push any block.)

Winning condition. The players win iff there exists a finite sequence of legal moves after which the set of block positions equals the set of goal positions:

$$\{B_1^{\text{final}},\dots,B_k^{\text{final}}\} = \{G_1,\dots,G_k\}.$$

(Equivalently, blocks can be assigned bijectively to goals at the end; labels do not matter.)

Communication problem. Alice's input and Bob's input are *identical*: the entire Sokoban instance

$$x = y = (W, H, S, (A_i)_{i=1}^k, (B_i)_{i=1}^k, (G_i)_{i=1}^k).$$

Define

$$f_{\text{Soko}}(x,y) = 1 \iff \text{the instance is solvable (a winning move sequence exists).}$$

Claim (Triviality). $D(f_{\text{Soko}}) = 0$ and $N^1(f_{\text{Soko}}) = N^0(f_{\text{Soko}}) = 0$.

Proof. Because $x = y$, both parties have the *entire* instance. With unbounded computation, each can decide solvability locally (e.g., by exhaustive search over the finite state space). Hence there is a protocol with zero communication: each outputs the computed answer. Thus $D(f_{\text{Soko}}) \le 0$, so $D(f_{\text{Soko}}) = 0$.

For nondeterminism, likewise communication is unnecessary: each party can ignore any certificate and compute the answer directly. So $N^1 = N^0 = 0$.

Tie-breaking discussion (not needed for solvability). If the task were instead to *execute* a solution without communication, the agents can still pre-agree on a deterministic rule to avoid symmetry (e.g., lexicographic priorities over agents/cells), because the full instance is common knowledge; but for the *decision* version, this is irrelevant.

Lower bound. Again, communication is always ≥ 0, so the bound is tight.

Embedding note. To force large communication, one must distribute the instance (e.g., each party sees disjoint subsets of walls/blocks/goals, or private observation transcripts). Under such partitions, one can embed DISJ or EQ by encoding bits into whether certain corridors/locks exist, producing $\Omega(n)$ or $\Omega(n^2)$ lower bounds. Without private inputs, the decision problem has 0 communication complexity.

References